

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
First Semester of M. Sc. (Mathematics) Examination November 2018
MA712 : Functions of Several Real Variables

Date : 27/11/2018, Tuesday

Time: 10:00 a.m. to 10:45 a.m.

Total marks 20

Multiple Choice Questions

Instructions:

1. Tick the correct answer only in this sheet.
2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
3. Maximum time to answer Multiple Choice Questions will be 45 minutes.
4. Each question carries one mark.
5. All notations and terminologies are standard.

Q-I Choose the correct answer from the given options in the followings:

[20]

- (1) Which of the following statement is true?
 - (a) $\pi_i(x) = x_i (x \in \mathbb{R}^n)$ is continuous at $a \in \mathbb{R}^n$, $(1 \leq i \leq n)$.
 - (b) $\pi_i(x) = x_i (x \in \mathbb{R}^n)$ is not continuous at $a \in \mathbb{R}^n$, $(1 \leq i \leq n)$.
 - (c) $f \pm g$ is not continuous at $a \in \mathbb{R}^n$, where $f, g: \mathbb{R}^n \rightarrow \mathbb{R}^m$ are continuous at $a \in \mathbb{R}^n$.
 - (d) None of these
- (2) Which of the following is the matrix corresponding to the linear transformation $T(x) = (2x_1, 3x_2, x_1)$, $x = (x_1, x_2) \in \mathbb{R}^2$.
 - (a) $\begin{bmatrix} 1 & 1 \\ 2 & 0 \end{bmatrix}$
 - (b) $\begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & 0 \end{bmatrix}$
 - (c) $\begin{bmatrix} 2 & 1 & 1 \\ 0 & 3 & 0 \end{bmatrix}$
 - (d) $\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$

- (3) If $x, y \in \mathbb{R}^n$, then _____.
- $\|x + y\|^2 + \|x - y\|^2 \neq 2\|x\|^2 + 2\|y\|^2$
 - $\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2$
 - $\|x + y\|^2 + \|x + y\|^2 = 2\|x\|^2 + 2\|y\|^2$
 - None of these
- (4) Let $A \subset \mathbb{R}^n, x \in \mathbb{R}^n$. If there exists $\delta > 0$ such that $x \in B(x, \delta) \subseteq A$, then x is called _____.
- Interior point of A
 - Exterior point of A
 - Boundary point of A
 - None of these
- (5) Let $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be differentiable at $a \in \mathbb{R}^n$ and let $g: \mathbb{R}^m \rightarrow \mathbb{R}^p$ be a differentiable at $f(a)$. Then _____.
- $D(g \circ f)(a) = Df(a) \circ Dg(f(a))$
 - $D(g \circ f)(a) = Dg(f(a)) \circ Df(a)$
 - $D(f \circ g)(a) = Dg(f(a)) \circ Df(a)$
 - None of these
- (6) Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ and $a, x, y, e_i \in \mathbb{R}^n$. Which of the following statement is not true?
- $D_{x+y}f(a) = D_xf(a) + D_yf(a)$
 - $D_{e_i}f(a) = D_if(a)$
 - $D_{sx}f(a) = sD_xf(a)$
 - None of these
- (7) Which of the following statements is not true for $f(x) = \begin{cases} 1 & ; \text{ if } x \in A \\ 0 & ; \text{ if } x \notin A \end{cases}$?
- where $A = \{x \in \mathbb{R}^2: x_1 > 0 \text{ and } 0 < x_2 < x_1^2\}$.
- A is open.
 - $D_xf(0)$ exists for all $x \in \mathbb{R}^2$
 - f is continuous at 0.
 - f is not continuous at 0.
- (8) If $f: \mathbb{R}^n \rightarrow \mathbb{R}$, such that $D_{i,j}f$ and $D_{j,i}f$ are continuous in some open subset of $\mathbb{R}^n$ containing a , then _____.
- $D_{i,j}f(a) = D_{j,i}f(a)$
 - $D_{i,j}f(a) \neq D_{j,i}f(a)$
 - $D_{i,j}f(a) = D_{j,i}f(a)$
 - None of these
- (9) Let $x, y \in \mathbb{R}^n$. Which of the following statement is not true?
- $|\langle x, y \rangle| \leq \|x\| \|y\|$
 - $\|x + y\| \leq \|x\| + \|y\|$
 - $\|x - y\| \leq \|x\| - \|y\|$
 - None of these
- (10) Which of the following statement is not true in $\mathbb{R}^n$?
- Every open ball is an open set.
 - Every closed ball is a closed set.
 - Each open rectangle is an open set.
 - None of these.
- (11) Which of the following statement is not true?
- $\lambda_l: \mathbb{R} \rightarrow \mathbb{R}$ is a Linear Transformation, where $\lambda_l(x) = lx$.
 - Let $\lambda: \mathbb{R} \rightarrow \mathbb{R}$ be a Linear Transformation, then $\exists$ unique $l \in \mathbb{R} \ni \lambda = \lambda_l$.
 - For $l \in \mathbb{R}$ and $f: \mathbb{R}^n \rightarrow \mathbb{R}$. Let $\lambda_l \circ f = lf$. Then λ_l is a Linear Transformation.
 - $\pi(x_1, x_2) = x_1, (x_1, x_2) \in \mathbb{R}^2$ is a Linear Transformation.
- (12) Which of the following function is not differentiable?
- $f(x) = (x_1^2 + x_2, x_3 + 2)$
 - $f(x) = x_1x_2x_3$
 - $f(x) = (x_1^2, x_2^2, x_1 + x_2)$
 - $f(x) = \begin{cases} \frac{x_1|x_2|}{\sqrt{x_1^2 + x_2^2}}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$
- (13) Which of the following statements is not true?
- $df(p)$ is not linear.
 - df is a 1-form on $\mathbb{R}^n$
 - $df(p)$ is linear.
 - $\widetilde{f}_{k*}: \Delta_{kF}(\mathbb{R}^m) \rightarrow \Delta_{kF}(\mathbb{R}^n)$ is a linear map.

- (14) Which of the following statements is correct?
- (a) $\mathcal{T}^l(V)$ is not a linear space. (b) The dimension of V and V^* are not same.
- (c) $\Lambda^k(V)$ is a linear subspace of $\mathcal{T}^k(V)$. (d) None of these
- (15) Let $V = \mathbb{R}^7$. Define $T: V \times V \rightarrow R$ by $T(x, y) = x_1y_2 + \sqrt{2}x_7y_1, x = (x_1, \dots, x_7), y = (y_1, \dots, y_7) \in V$. Which of the following statements is true?
- (a) $T \in \mathcal{T}^7(\mathbb{R}^2)$ (b) $T \in \mathcal{T}^1(\mathbb{R}^7)$
- (c) $T \in \mathcal{T}^2(\mathbb{R}^7)$ (d) None of these
- (16) Let $S, S_1, S_2 \in \mathcal{T}^k(V)$ and $T \in \mathcal{T}^l(V)$. Which of the following statements is not true?
- (a) $S \otimes T \in \mathcal{T}^{k+l}(V)$ (b) $(S_1 + S_2) \otimes T = S_1 \otimes T + S_2 \otimes T$
- (c) $(\alpha S) \otimes T = S \otimes (\alpha T) = \alpha(S \otimes T)$ (d) $(S_1 \cdot S_2) \otimes T = S_1 \otimes T \cdot S_2 \otimes T$
- (17) Which of the following is a basis element of $\Lambda^2(\mathbb{R}^3)$?
- (a) $\pi_2 \wedge \pi_1$ (b) $\pi_3 \wedge \pi_2$
- (c) $\pi_1 \wedge \pi_2$ (d) $\pi_1 \wedge \pi_1$
- (18) Which of the following is not a 2-tensor?
- (a) $\omega(x, y) = \frac{1}{2}(x_1y_2 - x_2y_1)$ on $\mathbb{R}^2$ (b) $\omega(x, y) = \frac{1}{2}(x_1y_3 - x_3y_1)$ on $\mathbb{R}^3$
- (c) $\omega(x, y) = (x_1y_2 - x_2y_1)$ on $\mathbb{R}^2$ (d) $\omega(x, y) = (x_1y_2 + x_3y_1)$ on $\mathbb{R}^3$
- (19) Let ω be k -form, η be l -form and $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable. Which of the following statements is not true?
- (a) $d(\omega + \eta) = d\omega + d\eta$ (b) $f^*(d\omega) = d(f^*\omega)$
- (c) $d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta$ (d) $d(\omega \wedge \eta) = (-1)^k d\omega \wedge \eta$
- (20) Which of the following statements is not true?
- (a) A singular 1-cube is often called a surface.
- (b) A finite sum of singular n -cubes with integer coefficients is called an n -chain in $A \subset \mathbb{R}^n$.
- (c) If c is an n -chain in A , then $\partial(\partial c) = 0$.
- (d) None of these.

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Date: 27/11/2018, Tuesday Time 10:45 a.m. to 1:00 p.m. Total marks of Section I & II 50

Instructions:

1. Section I and Section II must be written in separate answer books.
2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
3. Figures to the right indicate marks.
4. All notations and terminologies are standard.
5. V be a vector space over $\mathbb{R}$ with dimension n .

SECTION - I

Q-II Answer the followings:

(A) Attempt any *TWO* of the following:

[8]

- (1) Let $A \subset \mathbb{R}^n$ be closed, $f: A \rightarrow \mathbb{R}$ be a bounded function and $\varepsilon > 0$. Then show that the set $B = \{x \in A: 0(f; x) \geq \varepsilon\}$ is closed in $\mathbb{R}^n$.
- (2) State and prove implicit function theorem.
- (3) Let $(V, \langle \cdot, \cdot \rangle)$ be an n dimensional vector space over $\mathbb{R}$. Prove that each inner product T on V has an orthonormal basis. Also show that there is an isomorphism $f: \mathbb{R}^n \rightarrow V$ such that $f_2^*(T) = \langle \cdot, \cdot \rangle$.

(B) Attempt any *SIX* of the following:

[12]

- (1) If $x, y, z \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$, then prove that $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$ and $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle = \langle x, \alpha y \rangle$.
- (2) If $x, y \in \mathbb{R}^n$, then prove that $\langle x, y \rangle = \frac{1}{4} [\|x + y\|^2 - \|x - y\|^2]$.
- (3) Let $A \subset \mathbb{R}^n$, $f: A \rightarrow \mathbb{R}$ be a bounded function and $a \in A$. Show that if f is continuous at a , then $0(f; a) = 0$.
- (4) Let $p: \mathbb{R}^2 \rightarrow \mathbb{R}$. Define $p(x) = x_1 x_2$, $x = (x_1, x_2) \in \mathbb{R}^2$. Show that p is differentiable at a and $Dp(a)(x) = a_2 x_1 + a_1 x_2$.
- (5) If $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is a homogeneous function of degree m , then show that $f(x) = \frac{1}{m} \sum_{i=1}^n x_i D_i f(x)$.
- (6) Suppose $A \subset \mathbb{R}^n$ be an open set. If $f: A \rightarrow \mathbb{R}$ has maximum value at point $a \in A$, such that $D_i f(a)$ exists, then show that $D_i f(a) = 0$. Does the converse of above result true?
- (7) Let $f: V \rightarrow W$ be a linear map; $k, l \in \mathbb{N}$, $S \in \mathcal{T}^k(W)$ and $T \in \mathcal{T}^l(W)$. Show that $f_{k+l}^*(S \otimes T) = f_k^*(S) \otimes f_l^*(T)$.
- (8) Let $\omega \in \wedge^k(V)$, $\eta \in \wedge^l(V)$ and $\theta \in \wedge^m(V)$. Show that $(\omega \wedge \eta) \wedge \theta = \omega \wedge (\eta \wedge \theta) = \frac{(k+l+m)!}{k! l! m!} \text{Alt}(\omega \otimes \eta \otimes \theta)$.
- (9) Let ω be a k -form. Prove that $d(d\omega) = 0$.

SECTION – II

Q-III Answer the followings:

(A) Attempt any *THREE* of the following:

[18]

- (1) Define Linear transformation. Let $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$. Explain meaning of $\lim_{x \rightarrow a} f(x)$, $a \in \mathbb{R}^n$. Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation, then prove that there exists an $n \times m$ matrix $A = [a_{ij}]_{n \times m}$ such that $T(x) = xA$ ($x \in \mathbb{R}^n$).
- (2) Define differentiability of $f: \mathbb{R} \rightarrow \mathbb{R}$ and $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$. If a function $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable at $a \in \mathbb{R}^n$. Then show that there exists a unique linear transformation $\lambda: \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that
$$\lim_{h \rightarrow 0} \frac{\|f(a+h) - f(a) - \lambda(h)\|}{\|h\|} = 0.$$
- (3) State and prove chain rule on differentiability of $g \circ f$, where $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $g: \mathbb{R}^m \rightarrow \mathbb{R}^p$.
- (4) If a function $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is continuously differentiable at $a \in \mathbb{R}^n$, prove that $Df(a)$ exists.
- (5) Let $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$. Define partial derivative and directional derivative of f . If f is differentiable at $a \in \mathbb{R}^n$, show that the $(i, j)^{th}$ -entry of the Jacobian matrix $f'(a)$ is exactly the i^{th} -entry of the Jacobian matrix $(f^j)'(a)$, where $1 \leq i \leq n, 1 \leq j \leq m$.

(B) Attempt any *TWO* of the following:

[12]

- (1) State and prove Poincare lemma.
- (2) State and prove Stokes' theorem.
- (3) Define wedge product. Let V be an n dimensional vector space over $\mathbb{R}$. Show that the dimension of the vector space $\Lambda^k(V)$ is $\binom{n}{k}$.

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